Connect One



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https://github.com/Vegi-Jayakumar/Hackathon_DataCrusaders_ConnectOne

Problem Statement

ConnectOne is a regional telecom operator with 8 lakh prepaid and postpaid subscribers in Andhra Pradesh and Telangana. Every month thousands of subscribers move to other networks, and the company is not sure which plans or areas are losing the most customers.

Network complaints are rising in some towns, and many subscribers are on plans that are not profitable. Leadership wants to understand churn, usage and plan performance before revising plans and network investments.



Primary Responsibilities

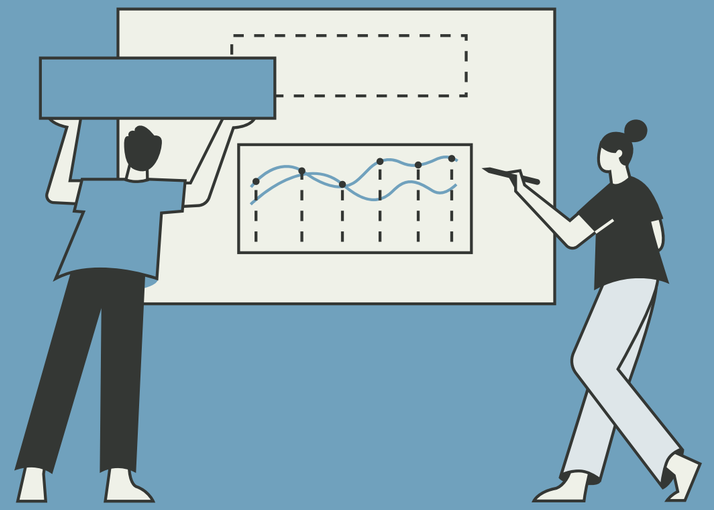
Collect Data | Validate | Analyze | Report | Recommend



Objective

1. Measure churn by plan, area and subscriber type
2. Find the usage and complaint patterns seen before subscribers leave
3. Identify profitable and loss-making plans
4. Find areas with poor network experience
5. Turn the findings into a dashboard and retention recommendations

Their analysis provides clarity in situations where uncertainty previously existed

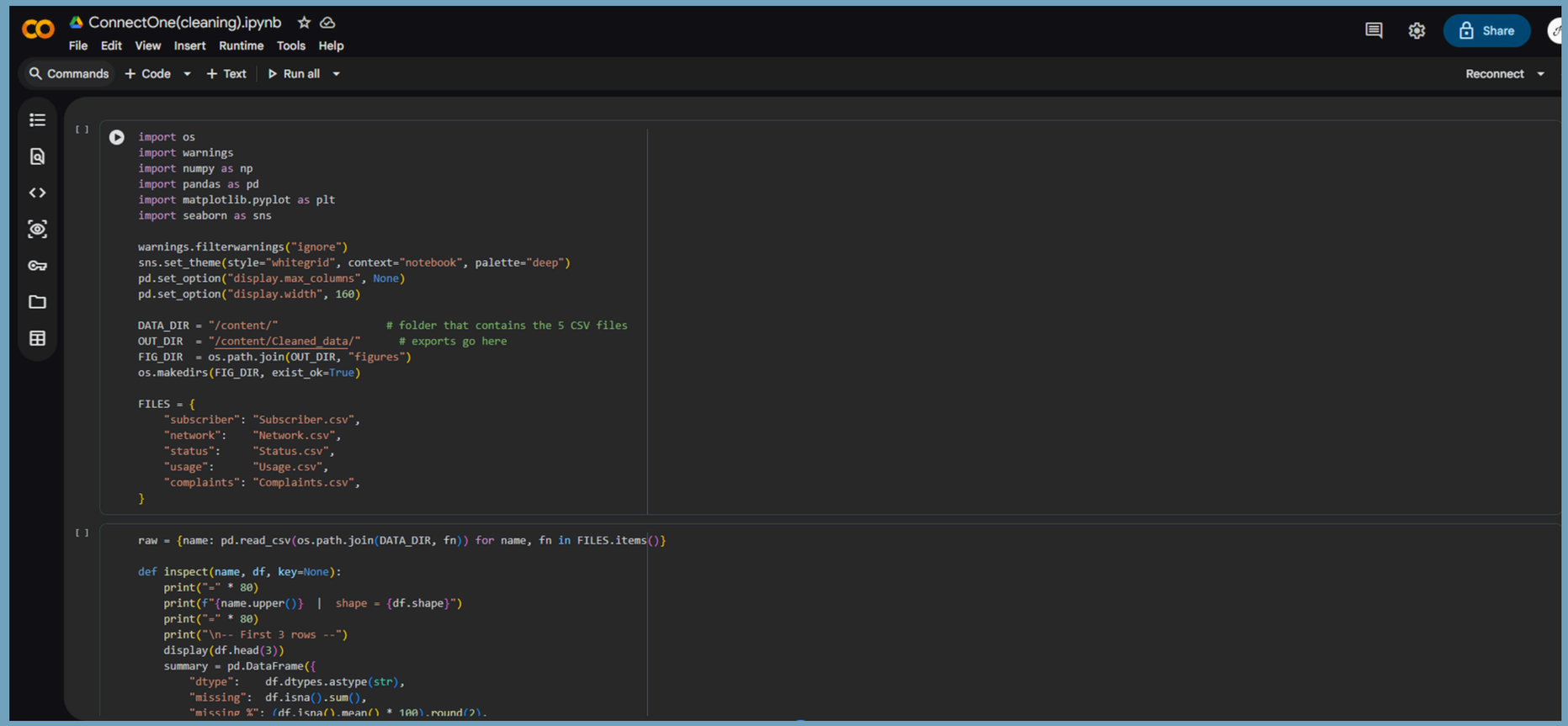


Data Collection

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	subscriber_id	activation_date	plan_type	plan_name	city	age_group	age	call_drops	avg_data_speed	tower_area	status	churn_date	ported_to	is_churned	month
2	SUB000000	2021-03-13	Prepaid	Standard	Tirupati	Adult	32	3	42.9	Mahbubnagar	Active		Not Applicable	0	November
3	SUB000001	2024-03-19	Postpaid	Basic	Visakhapatnam	Adult	62	0	124.73	Hyderabad	Active		Not Applicable	0	February
4	SUB000002	2021-11-25	Prepaid	Premium	Anantapur	Younger Adult	24	8	57.56	Mahbubnagar	Active		Not Applicable	0	January
5	SUB000003	2020-04-04	Prepaid	Premium	Kurnool	Older Adult	68	7	65.87	Kakinada	Active		Not Applicable	0	December
6	SUB000004	2023-10-30	Postpaid	Basic	Kurnool	Adult	34	7	136.99	Eluru	Active		Not Applicable	0	May
7	SUB000005	2022-12-28	Postpaid	Basic	Kakinada	Adult	46	4	93.84	Visakhapatnam	Active		Not Applicable	0	April
8	SUB000006	2020-12-24	Prepaid	Standard	Tirupati	Older Adult	60	3	121.39	Karimnagar	Churned	2021-05-11	Vodafone Idea	1	April
9	SUB000007	2020-09-23	Postpaid	Basic	Nalgonda	Older Adult	66	2	109.91	Mahbubnagar	Active		Not Applicable	0	March
10	SUB000008	2023-11-21	Prepaid	Standard	Guntur	Younger Adult	18	13	103.73	Hyderabad	Active		Not Applicable	0	December
11	SUB000009	2024-11-06	Postpaid	Basic	Suryapet	Younger Adult	23	1	119.94	Nalgonda	Active		Not Applicable	0	February
12	SUB000010	2020-01-21	Prepaid	Premium	Karimnagar	Older Adult	66	0	145.37	Nalgonda	Active		Not Applicable	0	November
13	SUB000011	2024-09-21	Prepaid	Basic	Vijayawada	Adult	30	2	108.69	Suryapet	Active		Not Applicable	0	December
14	SUB000012	2024-01-05	Postpaid	Standard	Visakhapatnam	Younger Adult	27	6	10.55	Nellore	Active		Not Applicable	0	September
15	SUB000013	2021-07-15	Postpaid	Premium	Guntur	Older Adult	61	7	42.37	Mahbubnagar	Active		Not Applicable	0	February
16	SUB000014	2022-10-08	Prepaid	Standard	Nellore	Older Adult	60	0	42.64	Guntur	Churned	2023-05-21	Jio	1	October
17	SUB000015	2023-09-09	Prepaid	Basic	Kurnool	Younger Adult	27	6	2.72	Nellore	Churned	2024-09-06	Jio	1	July
18	SUB000016	2023-10-15	Postpaid	Standard	Mahbubnagar	Adult	30	13	14.47	Hyderabad	Active		Not Applicable	0	January
19	SUB000017	2023-11-09	Prepaid	Basic	Siddipet	Adult	41	7	148.12	Kakinada	Active		Not Applicable	0	January
20	SUB000018	2024-05-05	Prepaid	Basic	Visakhapatnam	Older Adult	51	14	60.35	Anantapur	Active		Not Applicable	0	February
21	SUB000019	2021-03-27	Prepaid	Standard	Nalgonda	Adult	38	8	57.38	Nizamabad	Active		Not Applicable	0	April
22	SUB000020	2020-02-15	Postpaid	Basic	Nellore	Younger Adult	21	0	69.41	Rajahmundry	Active		Not Applicable	0	April
23	SUB000021	2020-07-13	Prepaid	Premium	Nalgonda	Younger Adult	20	5	104.92	Anantapur	Churned	2021-01-27	Airtel	1	September
24	SUB000022	2023-04-29	Prepaid	Premium	Karimnagar	Adult	44	13	70.56	Khammam	Active	0000-00-00	Not Applicable	0	October
25	SUB000023	2020-09-24	Prepaid	Basic	Kurnool	Adult	35	10	64.34	Vijayawada	Active	0000-00-00	Not Applicable	0	January
26	SUB000024	2021-06-23	Postpaid	Premium	Khammam	Younger Adult	24	8	97.25	Rajahmundry	Active	0000-00-00	Not Applicable	0	September
27	SUB000025	2023-03-25	Prepaid	Standard	Suryapet	Adult	39	4	116.04	Guntur	Active	0000-00-00	Not Applicable	0	April
28	SUB000026	2023-10-09	Postpaid	Standard	Anantapur	Adult	45	6	139.4	Ramagundam	Churned	2024-07-18	BSNL	1	December
29	SUB000027	2020-02-26	Postpaid	Basic	Visakhapatnam	Older Adult	73	10	56.01	Khammam	Churned	2021-04-09	Airtel	1	November
30	SUB000028	2024-10-03	Prepaid	Premium	Rajahmundry	Older Adult	65	3	70.05	Vijayawada	Active	0000-00-00	Not Applicable	0	December
31	SUB000029	2022-07-03	Postpaid	Basic	Karimnagar	Older Adult	72	2	30.91	Kakinada	Active	0000-00-00	Not Applicable	0	September
32	SUB000030	2021-07-26	Prepaid	Premium	Hyderabad	Adult	36	12	23.04	Hyderabad	Active	0000-00-00	Not Applicable	0	July
33	SUB000031	2022-11-10	Postpaid	Basic	Anantapur	Older Adult	64	3	97.33	Nellore	Churned	2024-02-03	BSNL	1	April
34	SUB000032	2022-12-22	Prepaid	Premium	Kakinada	Younger Adult	22	11	73.25	Nalgonda	Active	0000-00-00	Not Applicable	0	August
35	SUB000033	2020-12-27	Postpaid	Premium	Nellore	Older Adult	68	11	46.88	Kurnool	Active	0000-00-00	Not Applicable	0	October
36	SUB000034	2021-07-23	Postpaid	Basic	Hyderabad	Adult	47	8	115.27	Karimnagar	Active	0000-00-00	Not Applicable	0	May
37	SUB000035	2020-07-21	Postpaid	Premium	Tirupati	Adult	46	1	52.29	Suryapet	Active	0000-00-00	Not Applicable	0	January
38	SUB000036	2024-11-13	Prepaid	Standard	Guntur	Older Adult	57	14	66.4	Ramagundam	Active	0000-00-00	Not Applicable	0	March
39	SUB000037	2023-01-25	Postpaid	Premium	Nizamabad	Adult	39	3	50.35	Khammam	Active	0000-00-00	Not Applicable	0	December
40	SUB000038	2021-03-12	Postpaid	Standard	Tirupati	Younger Adult	25	12	5.6	Suryapet	Churned	2022-03-18	Airtel	1	July
41	SUB000039	2023-08-05	Postpaid	Basic	Warangal	Adult	36	2	113.47	Tirupati	Active	0000-00-00	Not Applicable	0	June
42	SUB000040	2024-08-13	Prepaid	Basic	Warangal	Older Adult	57	9	109.67	Visakhapatnam	Active	0000-00-00	Not Applicable	0	May
43	SUB000041	2021-03-19	Postpaid	Basic	Siddipet	Older Adult	53	11	38.09	Siddipet	Active	0000-00-00	Not Applicable	0	March



Data Cleaning



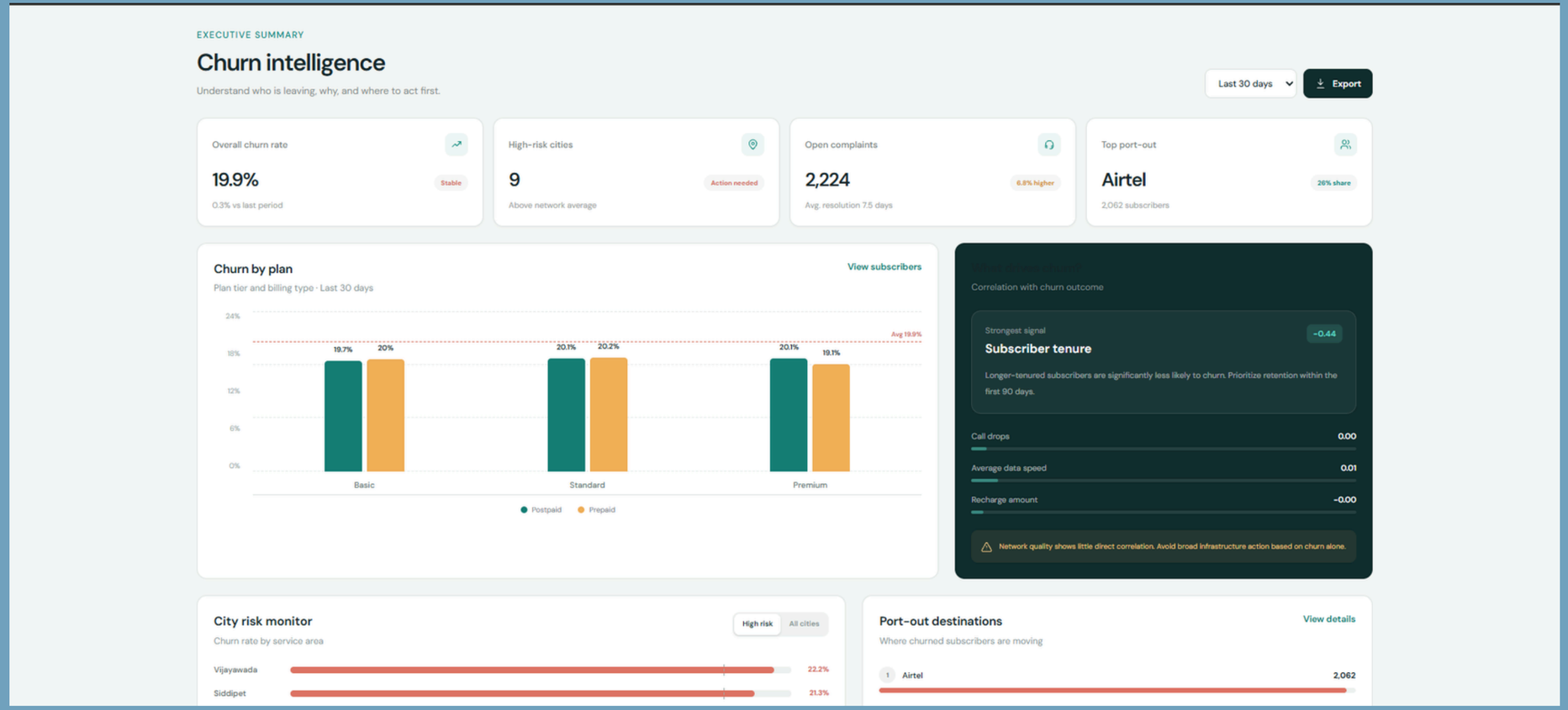
SQL Operations

```sql
CREATE DATABASE IF NOT EXISTS connectone;
USE connectone;

-- Re-runnable setup
DROP VIEW IF EXISTS vw_subscriber_summary;
DROP TABLE IF EXISTS complaints;
DROP TABLE IF EXISTS subscribers;

-- One row per subscriber. The master CSV already combines subscriber,
-- usage, network and status fields, so it is the main analysis table.
CREATE TABLE subscribers (
    subscriber_id       VARCHAR(20) PRIMARY KEY,
    activation_date     DATE,
    plan_type           VARCHAR(20),
    plan_name           VARCHAR(50),
    city                VARCHAR(80),
    age_group           VARCHAR(30),
    age                 INT,
    call_drops          INT DEFAULT 0,
    avg_data_speed_mbps DECIMAL(10,2),
    tower_area          VARCHAR(100),
    status              VARCHAR(20),
    ported_to           VARCHAR(50),
    is_churned          TINYINT(1) DEFAULT 0,
    month_name          VARCHAR(15),
    data_used_gb        DECIMAL(10,2),
    voice_minutes       INT,
    sms_count           INT,
    recharge_amount     DECIMAL(10,2),
    tenure_days         INT,
    CONSTRAINT chk_plan_type CHECK (plan_type IN ('Prepaid','Postpaid')),
    CONSTRAINT chk_status CHECK (status IN ('Active','Churned')),
    INDEX idx_plan_city (plan_name, city),
    INDEX idx_status (status),
    INDEX idx_tower_area (tower_area),
```

Dashboard


EXECUTIVE SUMMARY
Churn intelligence
Understand who is leaving, why, and where to act first.
Last 30 days
Export
Overall churn rate
19.9%
Stable
0.3% vs last period
High-risk cities
9
Action needed
Above network average
Open complaints
2,224
6.8% higher
Avg. resolution 7.5 days
Top port-out
Airtel
26% share
2,062 subscribers
Churn by plan
View subscribers
Plan tier and billing type · Last 30 days
24%
18%
12%
6%
0%
Avg 19.9%
19.7%
20%
20.1%
20.2%
20.1%
19.1%
Basic
Standard
Premium
Postpaid
Prepaid
Correlation with churn outcome
Strongest signal
-0.44
Subscriber tenure
Longer-tenured subscribers are significantly less likely to churn. Prioritize retention within the first 90 days.
Call drops
0.00
Average data speed
0.01
Recharge amount
-0.00
Network quality shows little direct correlation. Avoid broad infrastructure action based on churn alone.
City risk monitor
High risk
All cities
Churn rate by service area
Vijayawada
22.2%
Siddipet
21.3%
Port-out destinations
View details
Where churned subscribers are moving
1
Airtel
2,062

Insights

Longer-tenured subscribers are significantly less likely to churn.
Prioritize retention within the first 90 days.

Network quality shows little direct correlation.
Avoid broad infrastructure action based on churn alone.

Most Churn occurs due to VAS related Complaints.
Focus on improving the customer support on that front.

Thank You

Thank you for your attention

